DIGITAL LOGIC DESIGN PROJECT

Design of a 4-Bit Adder/Subtractor with Overflow Detection

Implemented and simulated in Logisim

Section 4

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Introduction and Objective

What is it?

An adder/subtractor is a combinational circuit that adds or subtracts two 4-bit binary numbers using one shared set of full adders.

A single control input K selects the operation: K = 0 adds, K = 1 subtracts using 2's complement.

An overflow flag tells us when the signed 4-bit result (range -8 to +7) is wrong.

Objectives

1. Design a 4-bit ripple-carry adder from full adders
2. Add subtraction using XOR gates and a mode input K
3. Detect signed overflow with $V = C3 \oplus C4$
4. Build and verify the circuit in Logisim

Theory and Logic Circuit

XOR as controlled inverter

B ⊕ K = B when K = 0 and B' when K = 1. K also feeds the first carry-in.

Add / Subtract

K = 0: S = A + B. K = 1: S = A + B' + 1 = A - B (2's complement).

Full adder (one bit)

S = A ⊕ B ⊕ Cin

Cout = A·B + Cin·(A ⊕ B)

Overflow detection

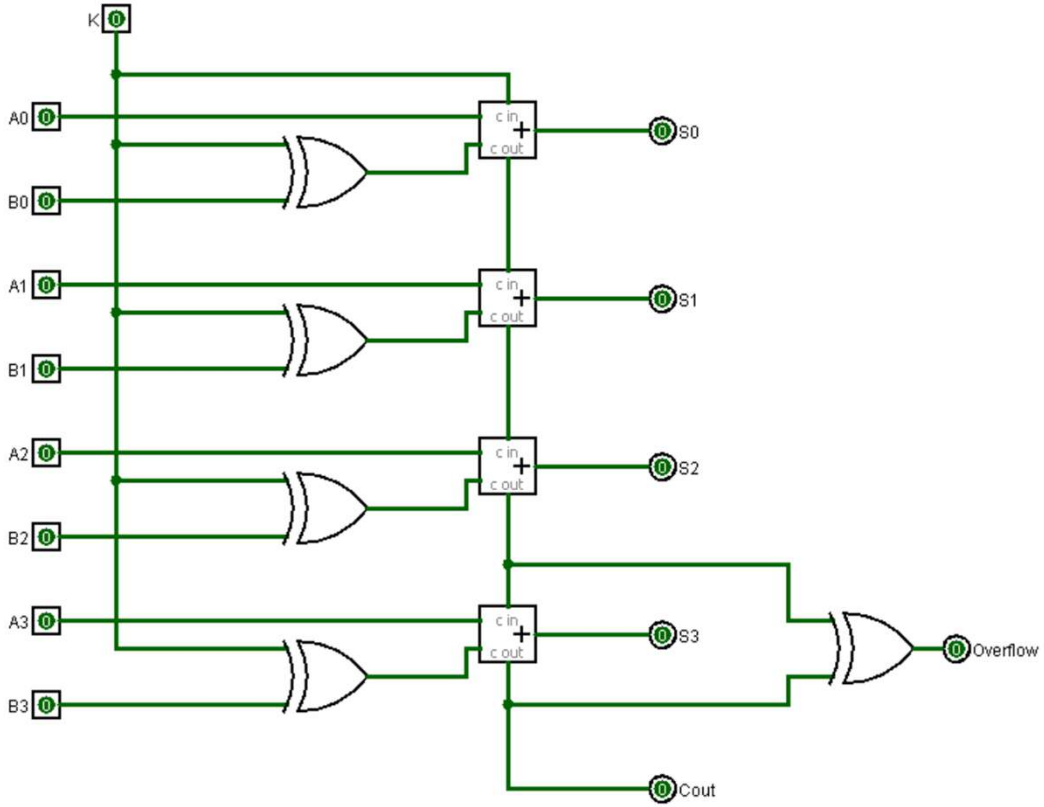
V = C3 ⊕ C4, the carry into the MSB XOR the carry out of the MSB.

Full adder truth table

A	B	Cin	S	Cout
0	0	0	**0**	**0**
0	0	1	**1**	**0**
0	1	0	**1**	**0**
0	1	1	**0**	**1**
1	0	0	**1**	**0**
1	0	1	**0**	**1**
1	1	0	**0**	**1**
1	1	1	**1**	**1**

Four of these full adders are chained: each Cout feeds the next Cin (ripple-carry).

Circuit Design in Logisim



1 Inputs
A3-A0, B3-B0 and the mode input K.

2 XOR stage
Each Bi is XORed with K, so B passes through (K = 0) or is inverted (K = 1).

3 Adder chain
Four full adders in ripple-carry. K drives the first carry-in, giving the +1 for subtraction.

4 Outputs
Sum bits S0-S3 and Cout. An extra XOR gate (carry into FA3 and Cout) drives the Overflow output.

Simulation Results and Truth Table

Op	A	B	K	S3 S2 S1 S0	Cout	C3	V	Signed result
Add	0101 (5)	0011 (3)	0	**1000**	0	1	**1**	8 > 7, overflow
Add	0010 (2)	0011 (3)	0	**0101**	0	0	**0**	5, correct
Add	1001 (-7)	1010 (-6)	0	**0011**	1	0	**1**	-13 < -8, overflow
Add	1100 (-4)	0011 (3)	0	**1111**	0	0	**0**	-1, correct
Sub	0111 (7)	0011 (3)	1	**0100**	1	1	**0**	4, correct
Sub	0011 (3)	0111 (7)	1	**1100**	0	0	**0**	-4, correct
Sub	0111 (7)	1010 (-6)	1	**1101**	0	1	**1**	13 > 7, overflow
Sub	1000 (-8)	0001 (1)	1	**0111**	1	0	**1**	-9 < -8, overflow

Overflow rule: V = C3 $\oplus$ Cout. V = 1 means the signed 4-bit result is invalid.

Subtraction: Cout = 1 means no borrow (A ≥ B as unsigned numbers).

Conclusion and Applications

Outcome

- One circuit performs both A + B and A - B
- K selects the mode; XOR gates plus carry-in do the 2's complement
- Overflow flag V correctly flags invalid signed results

Applications

- ALU of CPUs and microcontrollers
- Digital signal processing and calculators
- Address and counter arithmetic in embedded systems

Improvements

- Carry-lookahead adder for higher speed
- Extend to 8, 16 or 32 bits
- Add zero and negative flags for a full ALU

THANK YOU